Notes for week 4

- lists

a list holds many values in order

index starts at 0, [-1] is the last one

last valid index is len(labels) - 1

slicing [1:3], stop is not included

append and remove change the list

values[0] = 99 changes an item

for v in values loops over each one

IndexError = that position doesnt exist

list comprehensions

builds a new list in one line

[v * 2 for v in values]

[v for v in values if v >= 50] filters

tuples

like a list but cant be changed

point[0] = 5 gives TypeError

strings slice the same way, also locked

- dictionaries

  {"key": value}, read with d["key"]

missing key gives KeyError

.get("key", "n/a") is the safe way

d["status"] = "OK" adds or updates

loop keys, .values() or .items()

zip() builds a dict from two lists

- sets

  unique values only, set(list)

len(set(ips)) counts the unique ones

x in s checks if it is there

- files

with open(path) as f closes it for you

"w" writes, put \n on the end of a line

for line in f reads line by line

line.strip().split(",") gives the parts

- handling errors

try then except ValueError

name the error you expect

- list of dicts

the shape most real data comes in